Lesson 11 Teacher's Key: Wave After Wave

Unreal Engine (optional advanced track) - Instructor Reference Only

VIDEO GAME CREATION

What This Lesson Teaches

Extracting SpawnWave as its own Custom Event, called from both ResetGame and HandleEnemyDestroyed's nested Branch - continuing this course's Blueprint-native reuse pattern (UpdateHUD, SetStatus, ResetGame) into a fourth, closely related Custom Event.

Level 1 - Full Worked Solution

- SpawnWave is purely the enemy-spawning For Loop, extracted unchanged from what used to live inline inside ResetGame.
- ResetGame is now a thin wrapper: clear existing enemies, reset all variables including the new Wave, update the HUD, then Call SpawnWave.
- HandleEnemyDestroyed's nested Branch is where the real decision-making lives - Unreal's Blueprint equivalent of every other engine's if/else this week.

Why This Structure

- This mirrors Godot's spawn_wave() and Unity's SpawnEnemies() almost exactly, just expressed as a fourth Blueprint Custom Event alongside UpdateHUD, SetStatus, and ResetGame - worth naming this cross-engine parallel explicitly.
- Keeping the win/lose SetStatus messages reserved for the round's true end (not wave transitions) continues the same discipline established since Lesson 6.

Level 2 / Level 3 Talking Points

- For students who've built several Custom Events by now: ask them to sketch (verbally or on paper) the full call graph - BeginPlay -> ResetGame -> SpawnWave, and separately HandleEnemyDestroyed -> SpawnWave. A good comprehension check for how many Custom Events this project now has and how they relate.
- This is a good moment to ask advanced students what would need to change to make SpawnWave create a DIFFERENT number of enemies depending on Wave - not required this week (that's Lesson 14), but the nested Branch built today is exactly the hook a difficulty system would extend.

Common Student Mistakes

- Checking only EnemiesRemaining <= 0 without the nested Wave Branch.
- Calling SpawnWave from inside the True (win) case by mistake.

- Forgetting Wave = 1 inside ResetGame, leaving it at whatever value it had before a restart.